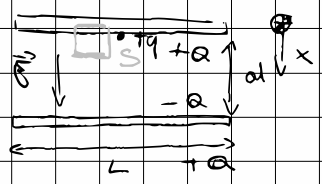
APPELLO 22 FEBBRAIO 2016

ES. 1

L, d $(d \ll L)$ $+Q$



a) $\pi R^2 E = \frac{Q}{\varepsilon_0 L^2} \pi R^2$

$\Rightarrow \vec{E} = \frac{Q}{\varepsilon_0 L^2} \vec{u}_x$

per $0 < x < d$

supposto lo spessore delle armature trascurabile.

$\vec{E} = 0$ per $x < 0$

$\vec{E} = \frac{Q}{\varepsilon L^2} \vec{u}_x$ per $x > d$

b) $\Delta V = \int_{V(0)}^{V(d)} dV = -\int_0^d \frac{Q}{\varepsilon_0 L^2} dx = -\frac{Qd}{\varepsilon_0 L^2}$

c) $U_{es} = \frac{QV}{2} = -\frac{1}{2} \frac{Q^2 d}{\varepsilon_0 L^2}$

d) $U_{es}^i = E_k^f + U_{es}^f$

$E_k^f = -\Delta U_{es} = \frac{1}{2} \frac{Q^2 d}{\varepsilon_0 L^2}$

$\Rightarrow v_f = \frac{Q}{L} \sqrt{\frac{d}{\varepsilon_0 m}}$

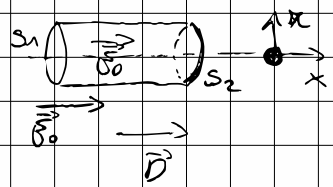
ES. 2:

a) $\nabla \times \vec{E} = 0 \qquad \nabla \cdot \vec{D} = \rho_{lib}$

con $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$\Rightarrow \quad [E_T] = 0 \qquad [D_n] = \rho_{lib}$

h; r cilindro ε_r $\quad \vec{E}_0 \parallel$ asse



b) · $h \ll r$

$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$

$[D_n] = 0 \quad \Rightarrow \quad [\vec{D}] = 0$

$\vec{E}_D = \dfrac{\vec{E}_0}{\varepsilon_r}$

· $h \gg r \qquad [E_T] = 0 \quad \Rightarrow \quad \vec{E}_D = \vec{E}_0$

c) · $h \ll r$

$\vec{D} = \varepsilon_0 \vec{E}_0 = \varepsilon_0 \vec{E}_D + \vec{P}$

$\Rightarrow \vec{P} = \varepsilon_0 \vec{E}_0 - \varepsilon_0 \dfrac{\vec{E}_0}{\varepsilon_r} = \varepsilon_0 \vec{E}_0 \left(\dfrac{\varepsilon_r - 1}{\varepsilon_r}\right)$

$\sigma_{pol} = \vec{P} \cdot \vec{n} \quad \Rightarrow \quad \sigma_{pol}^{S_1} = \dfrac{\varepsilon_0}{\varepsilon_r} E_0 (1 - \varepsilon_r)$

$\sigma_{pol}^{S_2} = \dfrac{\varepsilon_0}{\varepsilon_r} E_0 (\varepsilon_r - 1)$

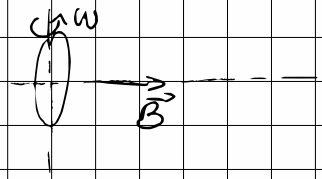
· $h \gg r$

$\vec{D} = \varepsilon_0 \vec{E}_0 = \varepsilon_0 \vec{E}_0 + \vec{P} \quad \Rightarrow \quad \vec{P} = 0$

$\Rightarrow \quad \sigma_{pol}^{S_1} = \sigma_{pol}^{S_2} = 0$

ES. 3:

R; a; $\vec{B}$



a) $\mathcal{E} = -\dfrac{d\Phi(\vec{B})}{dt}$

$$\Phi(\vec{B}) = \iint_S \vec{B}\cdot\vec{n}\,dS = \int_0^{2\pi}\int_0^a B\cos\theta\, r\,dr\,d\varphi =$$

$$= B\cos\theta\cdot\pi a^2$$

$$\frac{d\Phi(\vec{B})}{dt} = \frac{d}{dt}\left(B\cos\omega t\,\pi R^2\right) = -B\omega\pi a^2 \sin\omega t$$

$$\mathcal{E} = B\omega\pi a^2 \sin\omega t$$

$$P_{dissi} = \frac{\mathcal{E}^2}{R} = \left(B\omega\pi a^2 \sin\omega t\right)^2\cdot\frac{1}{R}$$

b) $\Phi_L(\vec{B}) = Li = L\dfrac{\mathcal{E}}{R}$

$$\frac{d}{dt}\left(\Phi(\vec{B}) + \Phi_L(\vec{B})\right) = \mathcal{E} + \frac{L}{R}\omega\mathcal{E}'$$

con $\mathcal{E}' = B\pi\omega a^2\cos\omega t$

$$P^*_{dissi} = \left(\mathcal{E} + \frac{L}{R}\omega\mathcal{E}'\right)^2\cdot\frac{1}{R} > P_{dissi}$$

ES. 4:

a) SIA $b >> a$ E $b >> \lambda$, L'EFFETTO DI DIFFRAZIONE È TRASCURABILE RISPETTO AL b.
POSTO INVECE $a \simeq \lambda$ SI PRESENTA DIFFRAZIONE LUNGO L'ASSE D'ESTENSIONE DI a.

ATTRAVERSO IL METODO DEI VETTORI ROTANTI SI OTTIENE:

$$\vec{E} = \vec{E}_0 \frac{\sin(\beta/2)}{\beta/2} \quad \text{CON} \quad \beta = k a \sin\theta = \frac{2\pi}{\lambda} n a \sin\theta$$

$$\Rightarrow \quad I = I_0 \frac{\sin^2(\beta/2)}{(\beta/2)^2}$$

b) $D = 1m$; $\lambda = 500nm$; $a = 10\mu m$; $b = 5mm$

$b >> a$; $b >> \lambda$ $\quad a \simeq \lambda$
SI PRESENTA DIFFRAZIONE SOLO LUNGO L'ASSE DI a

I MINIMI:

$$\frac{2\pi}{\lambda} a \sin\theta = 2\pi \qquad \sin\theta = \frac{2\lambda}{a}$$

$$\frac{x}{D} = \operatorname{tg}\theta \quad \Rightarrow \quad x = D \operatorname{tg}\left(\operatorname{arcsen}\left(\frac{2\lambda}{a}\right)\right) = 0{,}10m$$